CS 5402: Introduction to Data Mining

SP2026

(Lecture - 20)

Outlier Detection

Dr. Huiyuan Yang

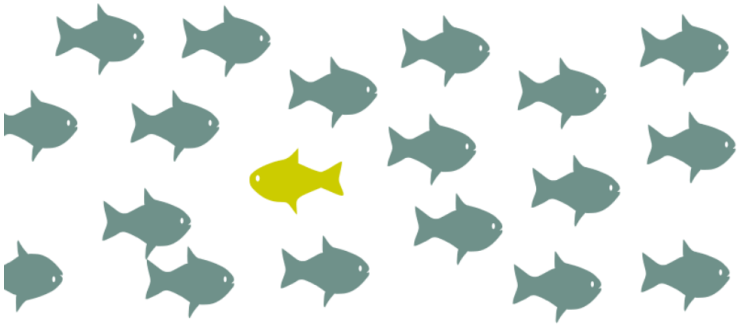
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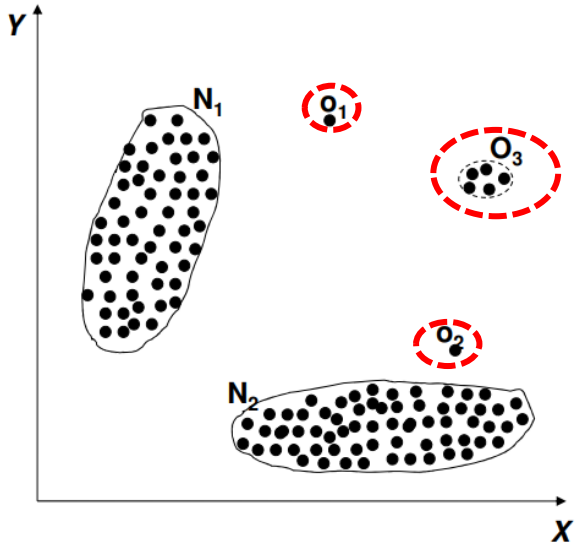
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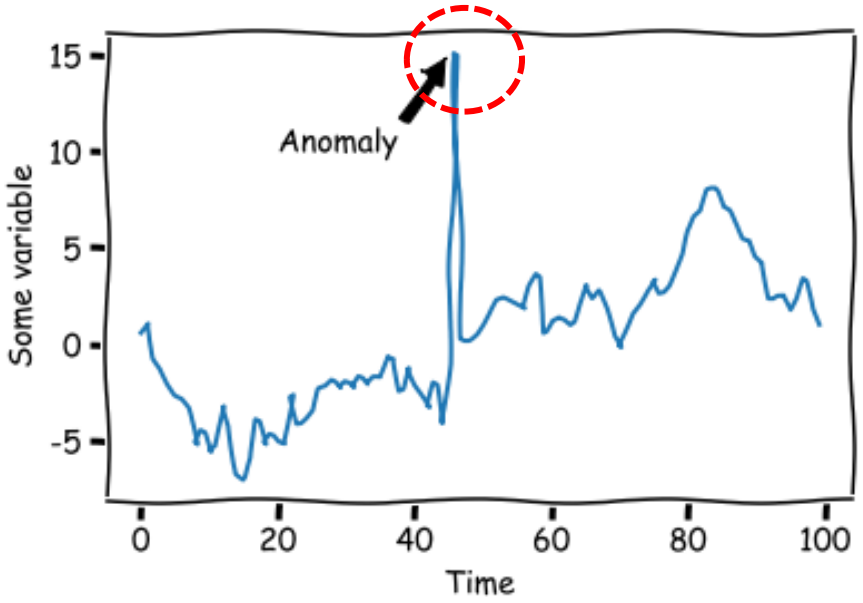
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Anomaly/Outlier Detection

- What are anomalies/outliers?
  - The set of data points that are considerably different than the remainder of the data







Anomaly/Outlier Detection

- Natural implication is that anomalies are relatively rare
  - One in a thousand occurs often if you have lots of data
  - Context is important, e.g., freezing temps in July
- Can be *important* or a *nuisance*
  - Unusually high blood pressure
  - 200 pound, 2 year old

Anomaly/Outlier Detection

- Anomaly detection (outlier detection or novelty detection) is referred to as the process of detecting data instances that significantly deviate from the majority of data instances.
- An active research area with the increasing demand and applications in broad domains, such as:
  - Risk management
  - Security
  - Financial surveillance
  - Health and Medical risk
  - AI safety
  - etc

Example of Application:

- Intrusion Detection;



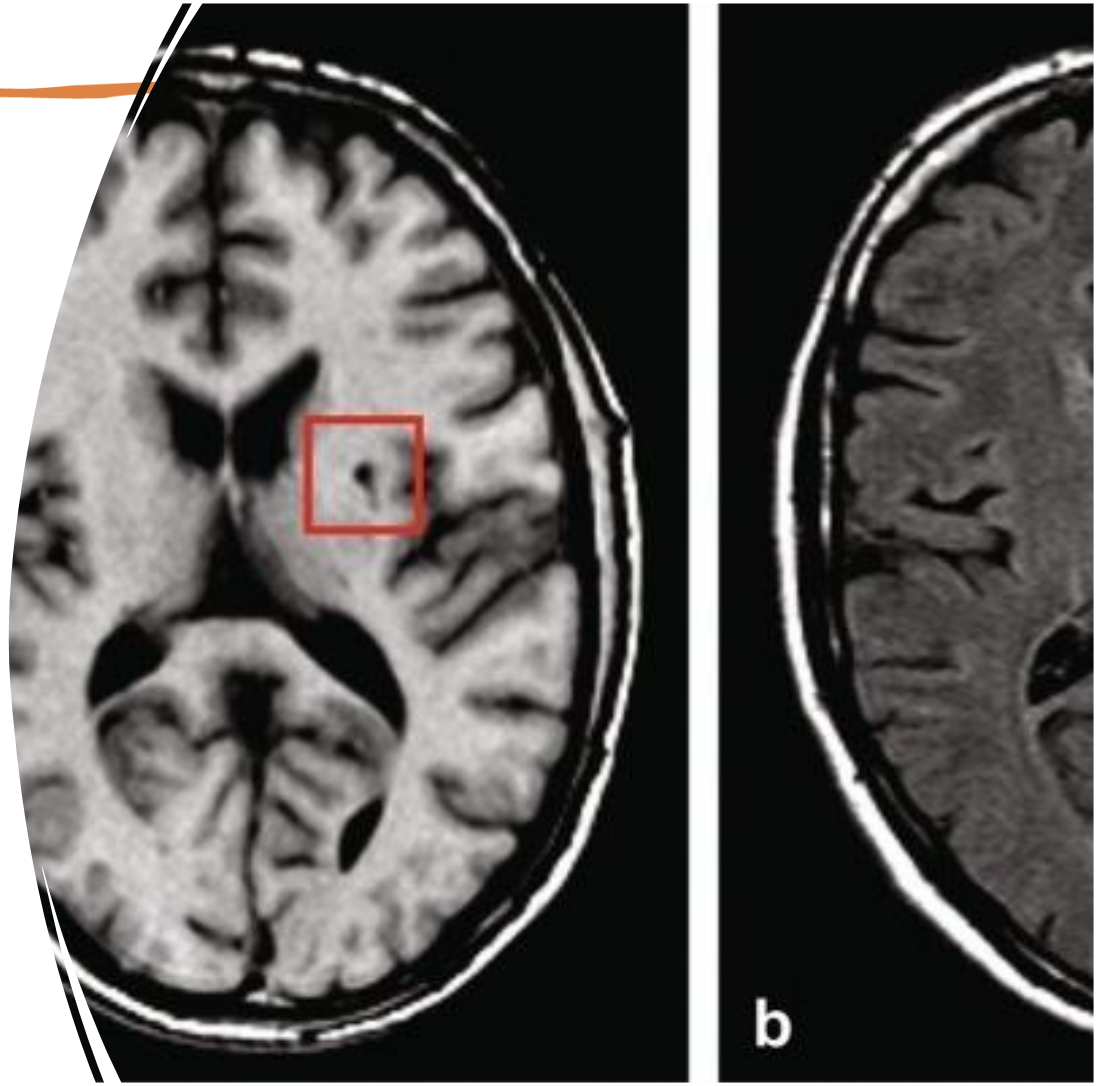
Example of Application:

- Fraud Detection;



Example of Application:

- Abnormal medical image detection;

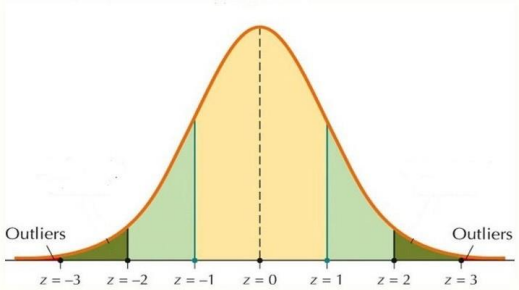


Causes of Anomalies

- Data from different classes
  - Measuring the weights of oranges, but a few grapefruit are mixed in



- Natural variation
  - Unusually tall people



- Data errors
  - 200 pound 2 year old

Distinction Between Noise and Anomalies

Noise

- Random fluctuations or measurement errors
- **No underlying pattern or significance**
- Often caused by:
  - Sensor errors
  - Data corruption
  - Environmental randomness

Example:

- Slight variation in ECG signal due to sensor noise

Typically **ignored or smoothed out**

Noise = random error
Anomaly = meaningful deviation

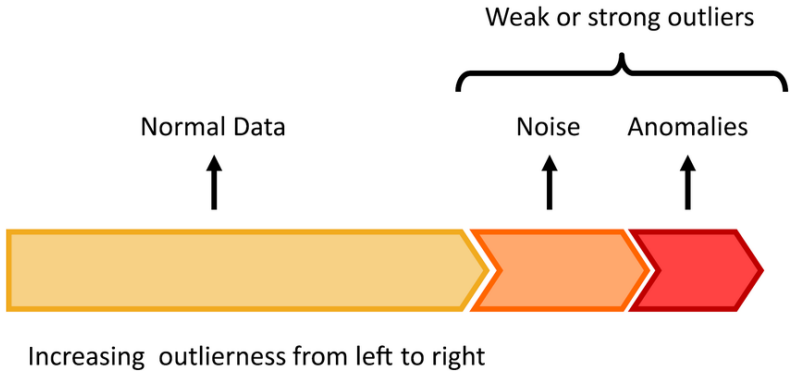
Anomalies (Outliers)

- Data points that **deviate significantly from normal behavior**
- Often **meaningful and informative**
- May indicate:
  - Rare events
  - System faults
  - Medical abnormalities

Example:

Sudden abnormal spike in heart rate → possible arrhythmia

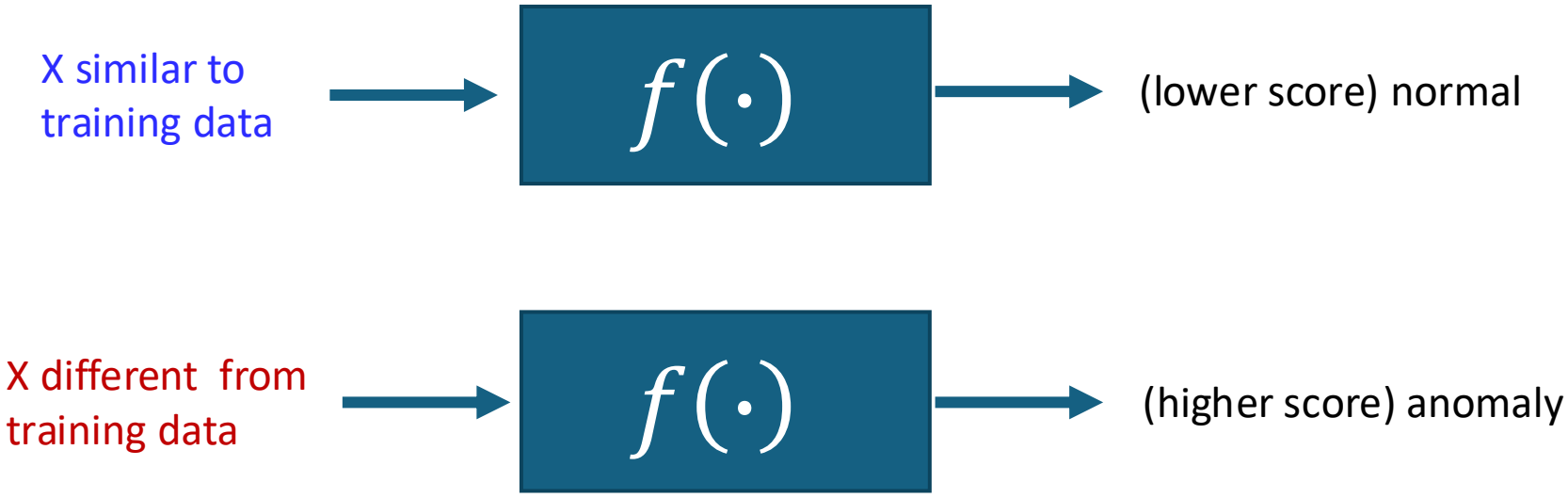
Important for detection and analysis



Problem Formulation

Given a dataset: $\mathrm{X} = \{\mathrm{x}_1, \mathrm{x}_2, \ldots, \mathrm{x}_\mathrm{N}\}$

An anomaly score learning function: $f(\cdot)$: $\mathrm{x} \rightarrow \mathbb{R}$



Decision Rule: x is an outlier if f(x)>τ, τ: threshold

Three Learning Settings (1)

Supervised:

- Data: normal + labeled outliers $D = (x_i, y_i),\ \ y_i \in \{normal, anomaly\}$
- **Objective:** Learn a classification function:

$$f(x) \rightarrow \{0, 1\}$$

Advantages

- Can leverage standard classification models (e.g., SVM, neural networks)
- Typically achieves strong performance when sufficient labeled data is available

Limitations

- Anomalies are **rare and hard to label**
- Difficult to cover all types of anomalies
- Severe class imbalance

Example

- Fraud detection with labeled transactions
- Medical datasets with confirmed abnormal cases

Three Learning Settings (2)

Semi-Supervised (Most common):

- Only **normal data** is available during training.

$$D_{train} = x_i,\ x_i \sim normal$$

- **Objective:** Learn a model of normality:

$$f(x) = normality\ score$$

- Detection rule:
  - *x is an anomaly if $f(x)$ is low (or high)*

Common Methods

- One-Class SVM
- Autoencoders
- Density estimation
- Self-supervised learning

Advantages

- Does not require anomaly labels
- Matches many real-world scenarios (e.g., healthcare, industrial systems)

Limitations

- Assumes training data is mostly clean
- Sensitive to distribution shifts

Example

- Modeling normal ECG signals to detect arrhythmia
- Learning normal machine behavior to detect faults

Three Learning Settings (3)

Unsupervised:

- No labels at all: $D = \{x_1, x_2, \ldots, x_n\}$

Core Assumptions: anomalies are:

1) Rare
2) Different from the majority

Objective: Learn an outlier scoring function:

$$f(x) = outlier\ score$$

Based on:

- Distance
- Density
- Local structure

Advantages

- No labeled data required
- Broad applicability

Limitations

- Assumptions may not always hold
- May misclassify rare but normal points as anomalies

Example

- Detecting anomalies in unlabeled system logs
- Identifying unusual patterns in network traffic

Key Assumptions

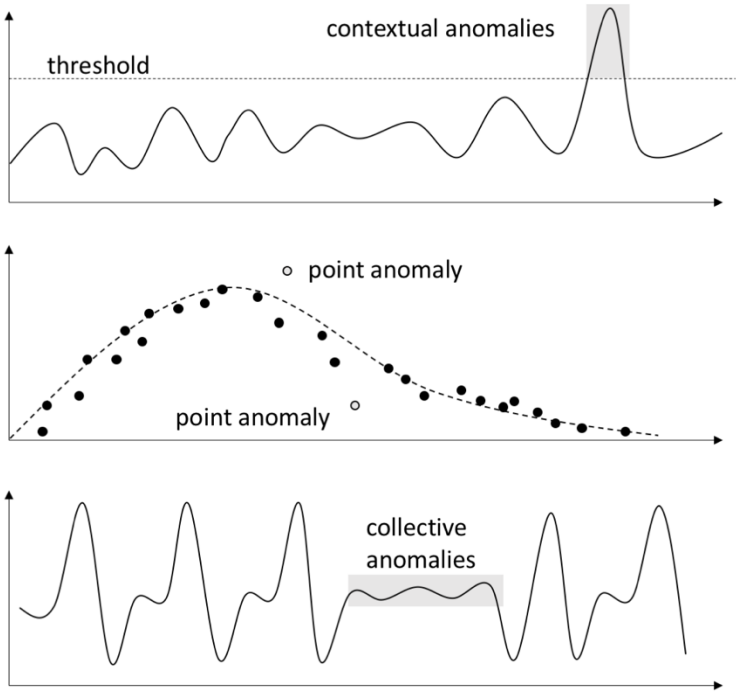
Outlier detection typically relies on:

- **Rarity**
  → anomalies are few
- **Different behavior**
  → statistically distinguishable
- **Consistency of normal data**
  → normal patterns are learnable

If these fail → detection becomes difficult

Types of Outliers

- **Point Anomalies**
  - Single data point is abnormal
- **Contextual Anomalies**
  - Abnormal in a specific context (e.g., heart rate during sleep)
- **Collective Anomalies**
  - A group of points is abnormal (e.g., unusual sequence pattern)



Anomaly Detection Techniques

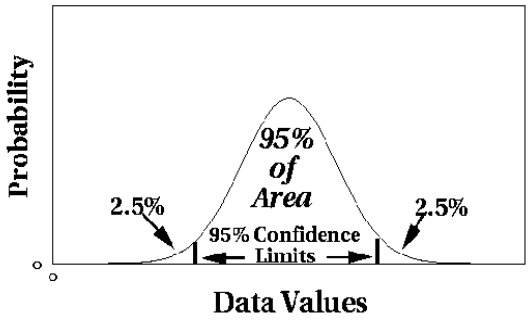
- Statistical Methods
  - Z-Score/Standard Score
  - IQR
- Distance-Based Methods
  - K-Nearest Neighbors (KNN)
  - Local Outlier Factor (LOF)
  - DBSCAN (Density-Based Spatial Clustering)
- Cluster-Based Methods
  - K-Means
  - Gaussian Mixture Models
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- Time-Series Anomaly Detection
  - Moving Average
  - LSTM Autoencoder / RNNs

Statistical Methods

Probabilistic definition of an outlier:

An outlier is an object that has a low probability with respect to a probability distribution model of the data.

- Usually assume a parametric model describing the distribution of the data (e.g., normal distribution)
- Apply a statistical test that depends on
  - Data distribution
  - Parameters of distribution (e.g., mean, variance)
  - Number of expected outliers (confidence limit)

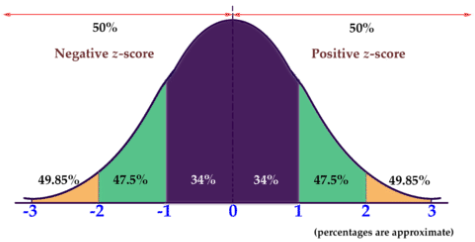


Statistical Methods

- Z-Score/Standard Score:
  - Z-score tells us how many standard deviations a value is from the mean.
  - Rule: A data point is considered an outlier if:

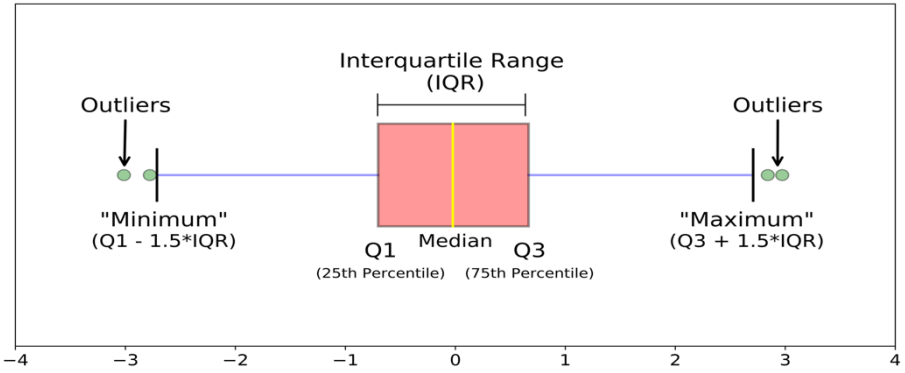
$$Z - score = \frac{\mathrm{x} - \mu}{\sigma}$$

Z-score < -3 or Z-score > 3



Statistical Methods

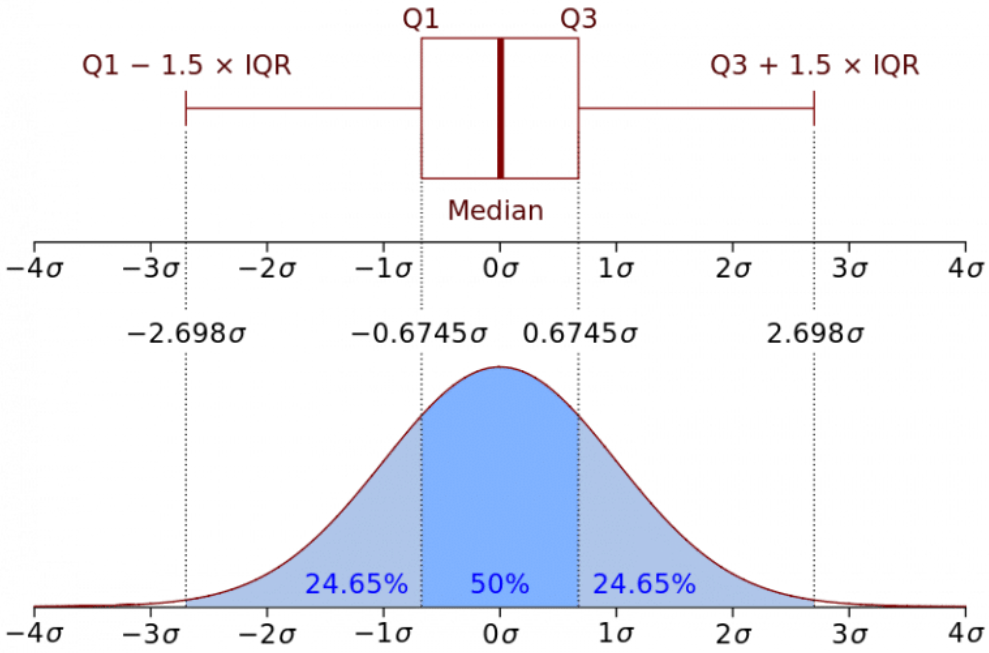
- IQR (Interquartile Range)



$IQR = Q_3 - Q_1$

Lower Outlier $= Q_1 - 1.5 \times IQR$

Higher Outlier $= Q_3 + 1.5 \times IQR$



Strengths/Weaknesses of Statistical Approaches

Core Idea

- Model the data distribution and identify points with low probability

Strengths

- Strong theoretical foundation
  → grounded in probability and statistics
- Computationally efficient
  → especially for simple distributions (e.g., Gaussian)
- High accuracy when assumptions hold
  → performs well if the true distribution is known

Limitations

- Distribution often unknown in practice
  → real-world data rarely follows ideal assumptions
- Struggles with high-dimensional data
  → estimating joint distributions becomes difficult
- Sensitive to anomalies
  → outliers can distort parameter estimation (mean, variance)

Anomaly Detection Techniques

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Distance-Based Approaches

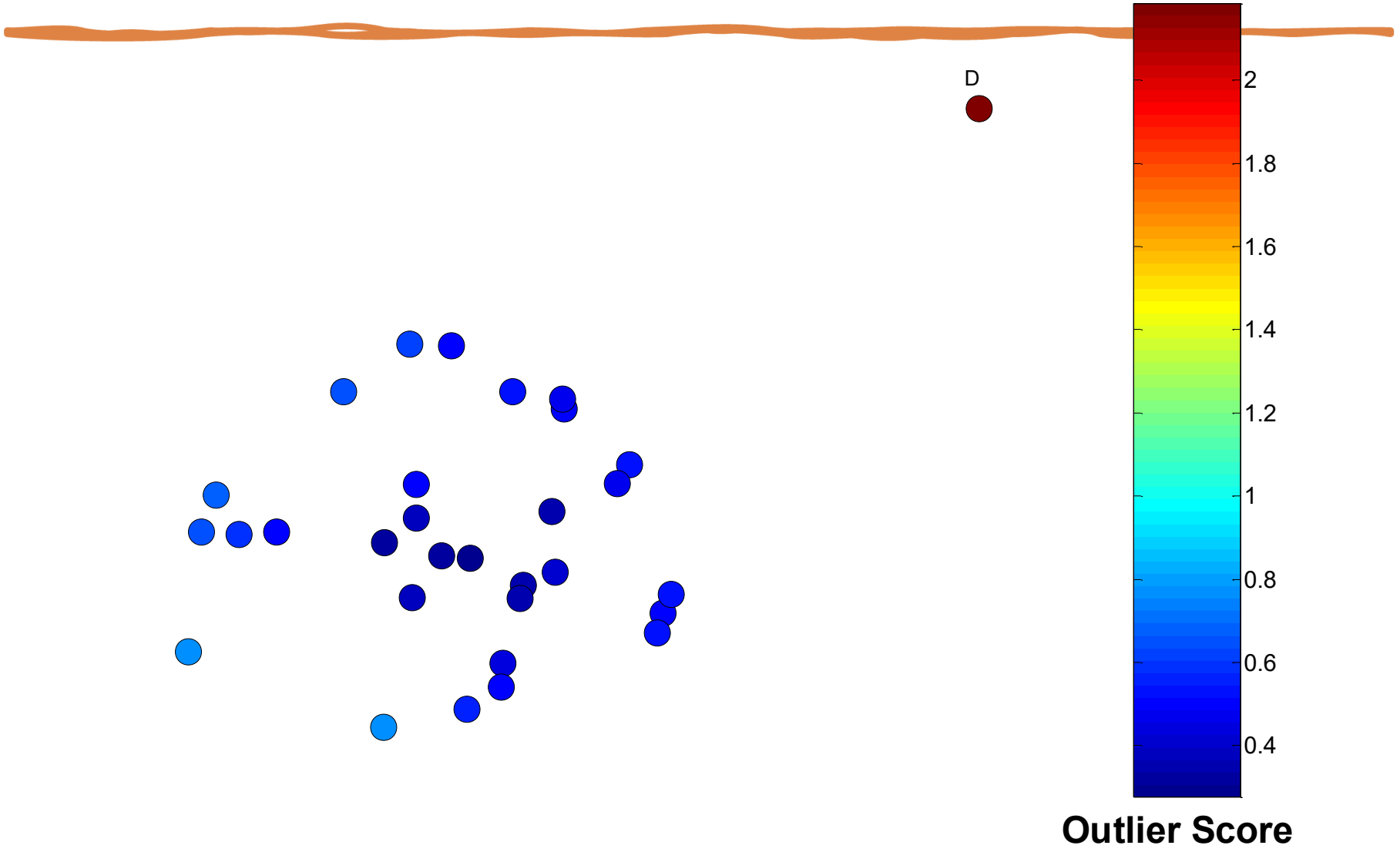
Outliers are far away from their nearest neighbors:

- For each pint, calculate its distance to the K nearest neighbors
- Points with a large average distance are likely outliers.

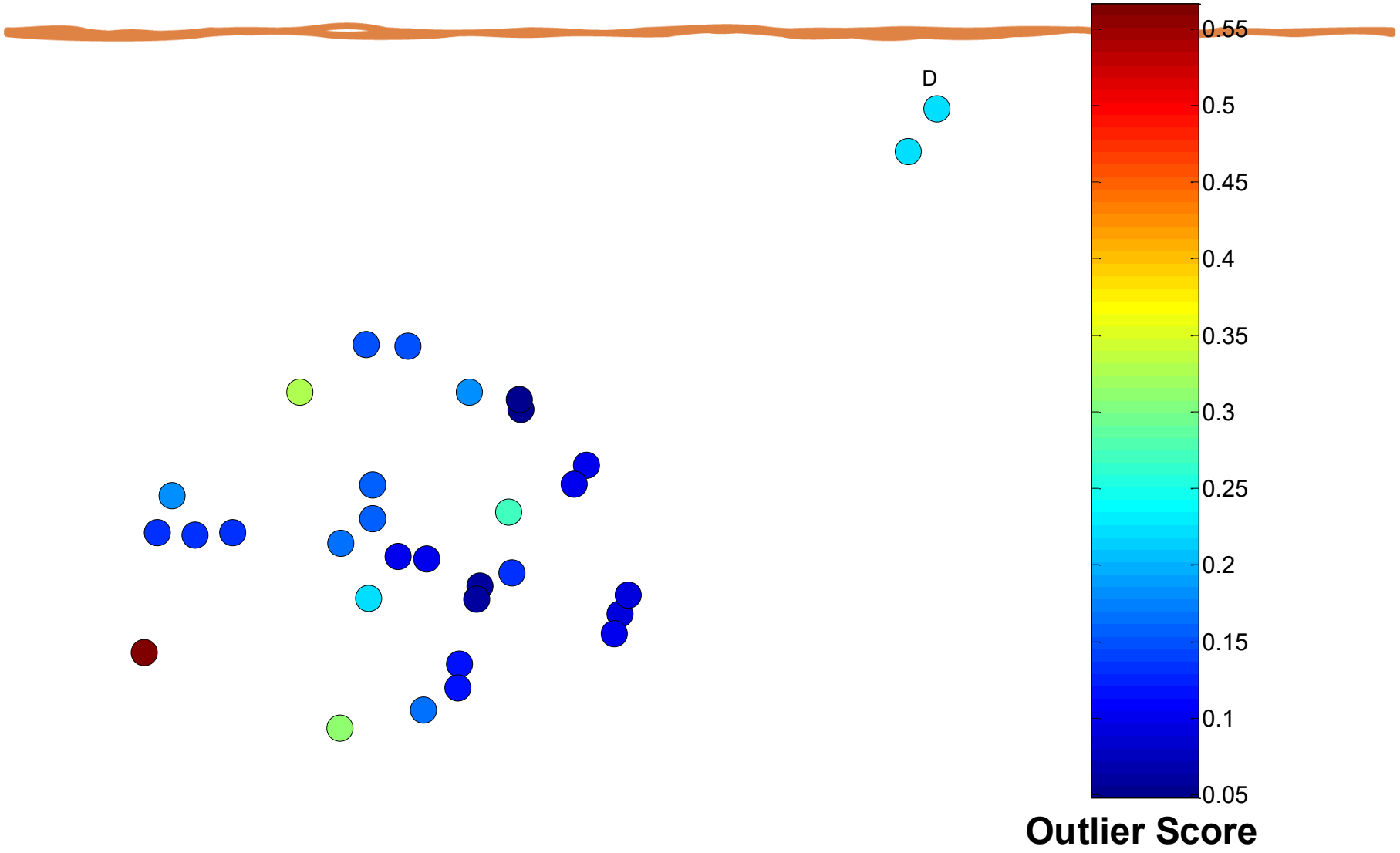
OR

- Count the neighborhood of an object based on a given radius.
- An object is then considered an outlier if its neighborhood does not have enough other points.

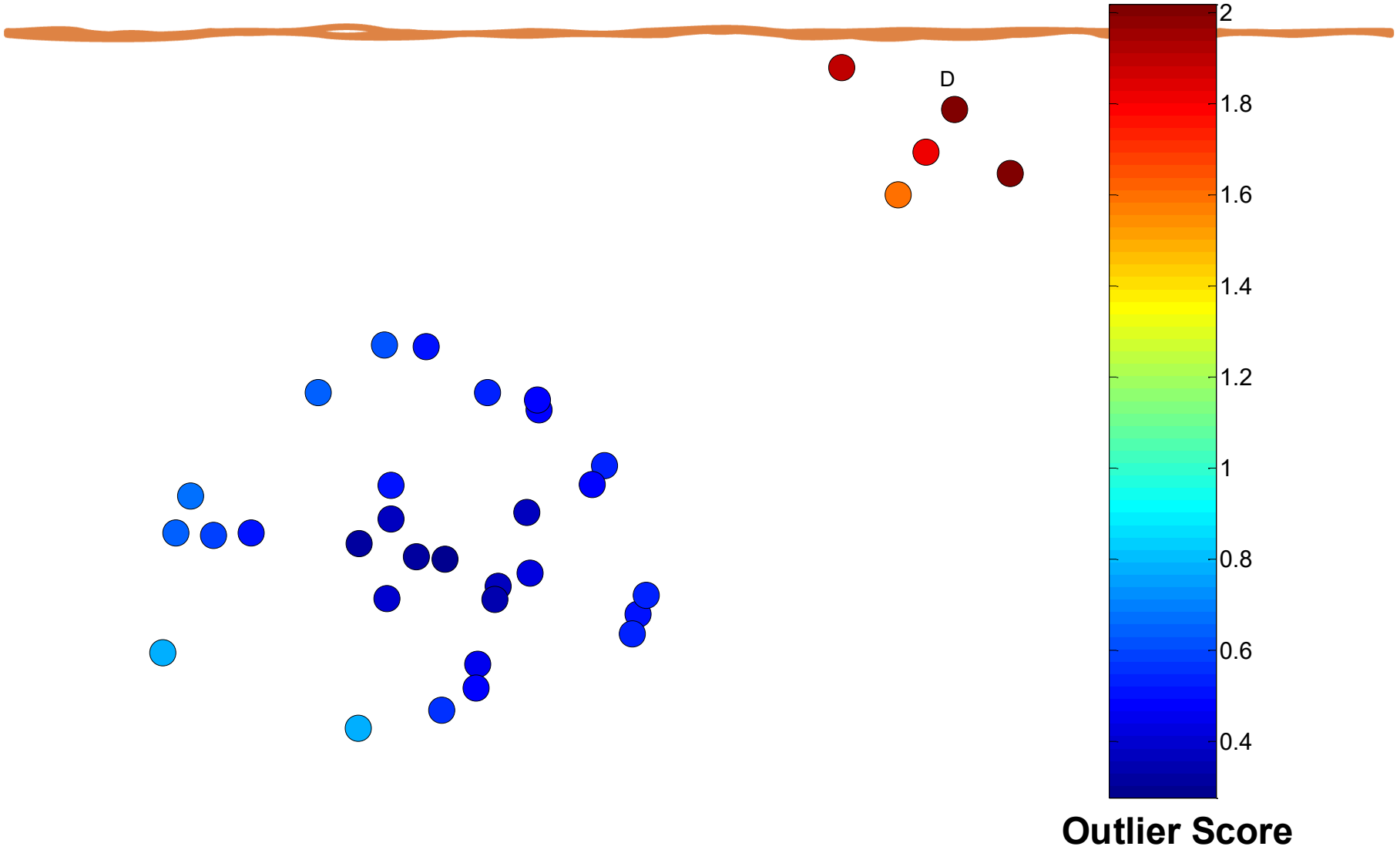
One Nearest Neighbor - One Outlier



One Nearest Neighbor - Two Outliers



Five Nearest Neighbors - Small Cluster

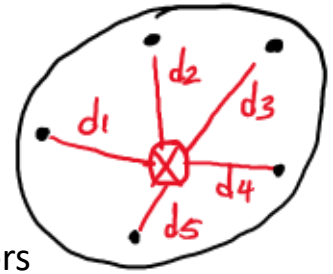


Distance-Based Approaches: LOF

The Local Outlier Factor (LOF) is used for outlier detection by measuring the local density deviation of a data point relative to its neighbors. A data point is identified as outlier if its density is significantly lower than its surroundings.

Steps:

- Choose K neighbors
  - For each data point, find its K nearest neighbors
- Compute local reachability density (LRD):
  - Measures how dense the neighborhood is around a point.
  - It's the inverse of the average distance to neighbors.
- Compare densities:
  - For each point, compare its LRD to those of its neighbors
- Compute LOF score:
  - ≈ 1 means the points' density is similar to its neighbors (Normal point)
  - >> 1 means the point is in a sparser region than neighbors (Outlier)
  - << 1 (rare) can mean a dense region inside a sparse one (depends on data structure)



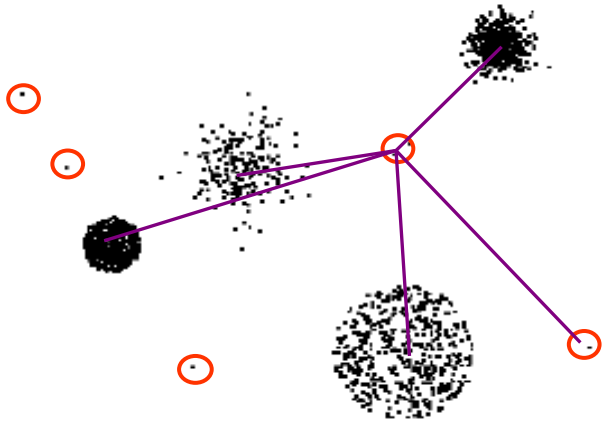
Anomaly Detection Techniques

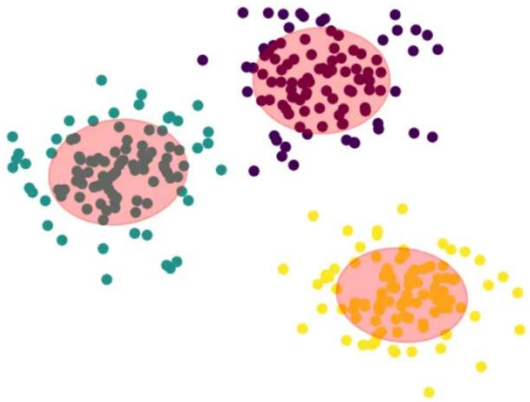
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Clustering-Based Approaches

An object is a cluster-based outlier if it does not strongly belong to any cluster

- For prototype-based clusters, an object is an outlier if it is not close enough to a cluster center
  - Outliers can impact the clustering produced
- For density-based clusters, an object is an outlier if its density is too low
  - Can't distinguish between noise and outliers
- For graph-based clusters, an object is an outlier if it is not well connected





Strengths/Weaknesses of Clustering-Based Approaches

Core Idea: Group similar data points into clusters; points that do not belong well to any cluster are considered outliers

Strengths

- Conceptually simple and intuitive
  → easy to explain and understand
- Flexible
  → can leverage many clustering algorithms (e.g., k-means, DBSCAN, hierarchical)
- No labeled data required
  → naturally fits unsupervised settings

Limitations

- Choice of clustering algorithm is non-trivial
  → different methods produce very different results
- Need to determine number of clusters (k)
  → often unknown and hard to tune
- Sensitive to outliers
  → anomalies can distort cluster centers or boundaries

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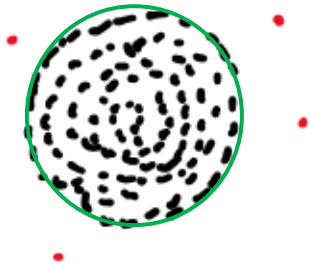
Machine Learning-Based Methods: One-Class SVM

One-Class Support Vector Machine (One-Class SVM) is an unsupervised algorithm that learns the boundary of "normal" data and identifies outliers or novelties as points that fall outside that boundary.

Support Vector Data Description (SVDD) method tries to:

Find the smallest possible sphere that encloses most of the data.

- Normal points lie insider the sphere
- Outliers lie outside the sphere.



Reconstruction-Based Approaches

- Based on assumptions there are patterns in the distribution of the normal class that can be captured using lower-dimensional representations

- Reduce data to lower dimensional data
  - E.g. Use Principal Components Analysis (PCA) or Auto-encoders

- Measure the reconstruction error for each object
  - The difference between original and reduced dimensionality version

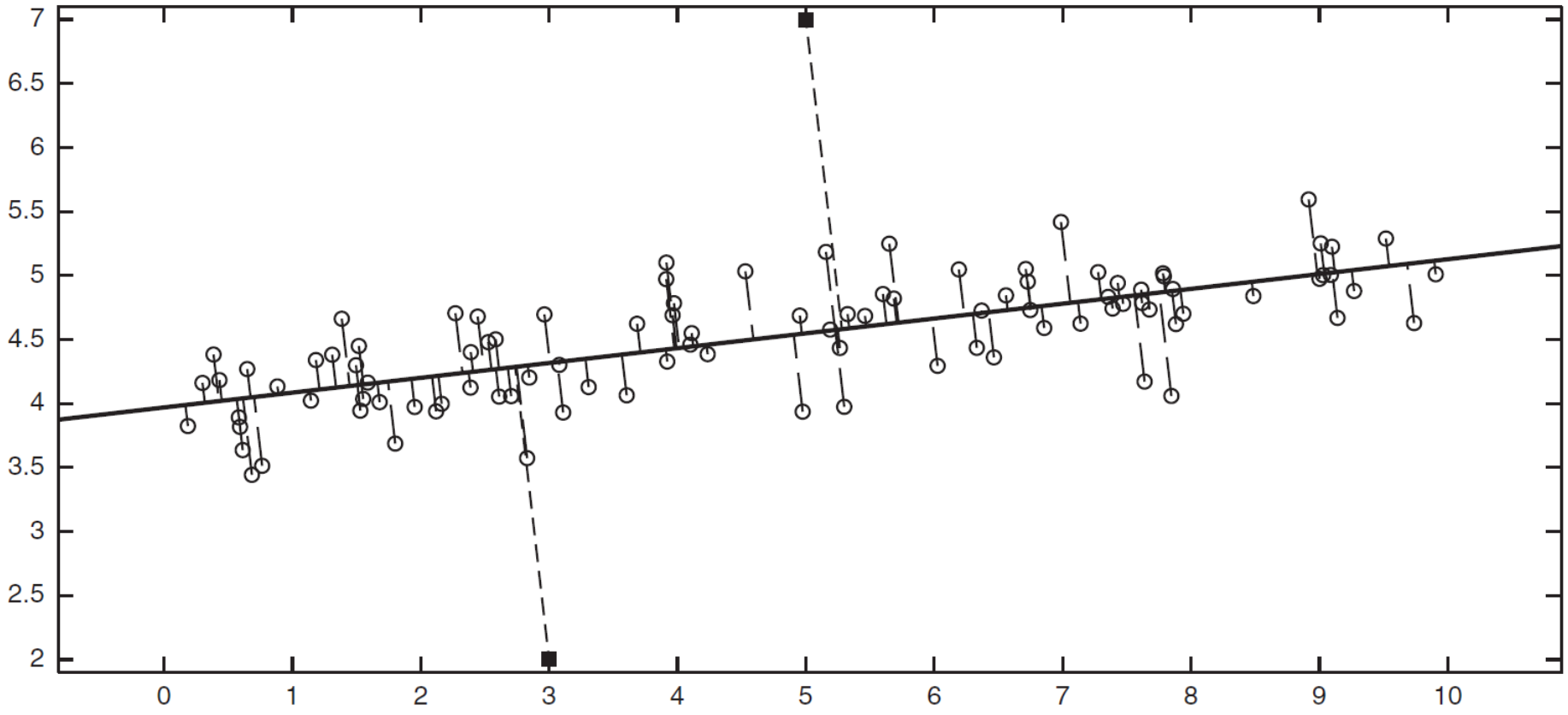
Reconstruction Error

- Let $\mathbf{x}$ be the original data object
- Find the representation of the object in a lower dimensional space
- Project the object back to the original space
- Call this object $\hat{\mathbf{x}}$

$$\text{Reconstruction Error}(\mathbf{x}) = \|\mathbf{x} - \hat{\mathbf{x}}\|$$

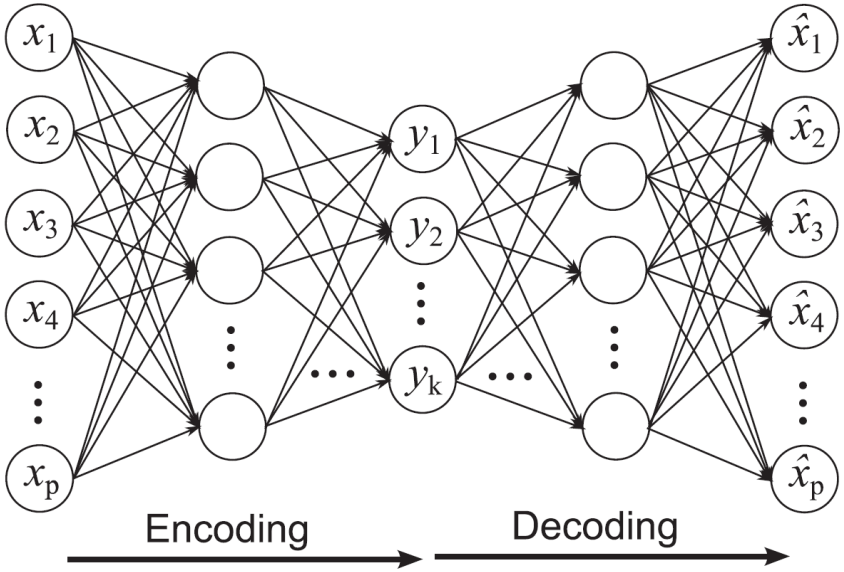
- Objects with large reconstruction errors are anomalies

Reconstruction of two-dimensional data



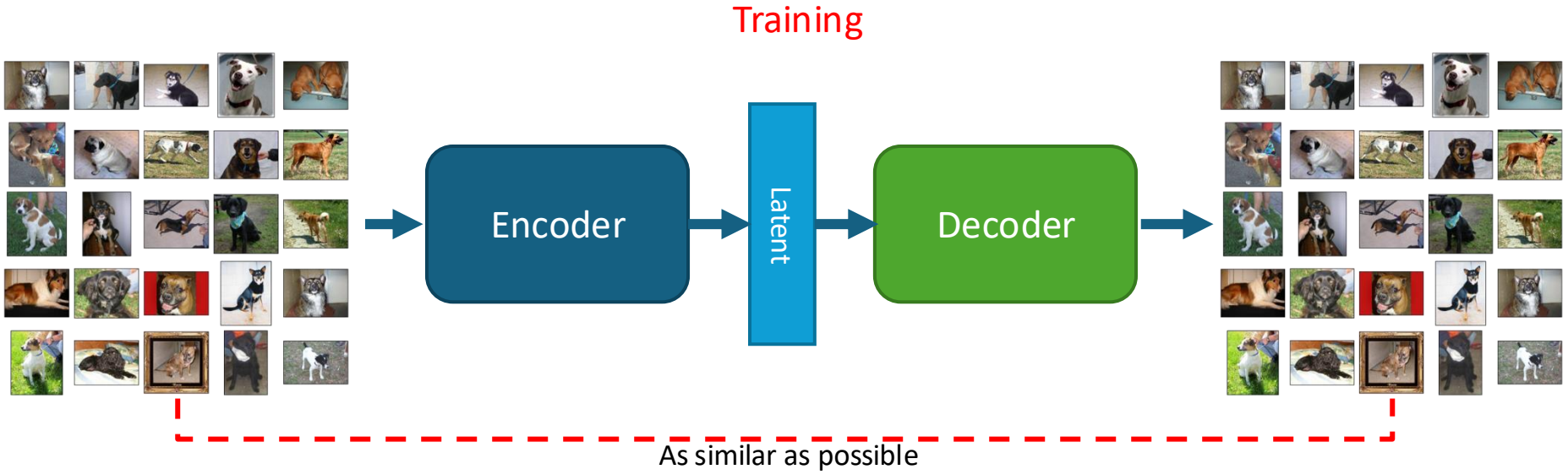
Basic Architecture of an Autoencoder

- An autoencoder is a multi-layer neural network
- The number of input and output neurons is equal to the number of original attributes.



Methods: *Data reconstruction*

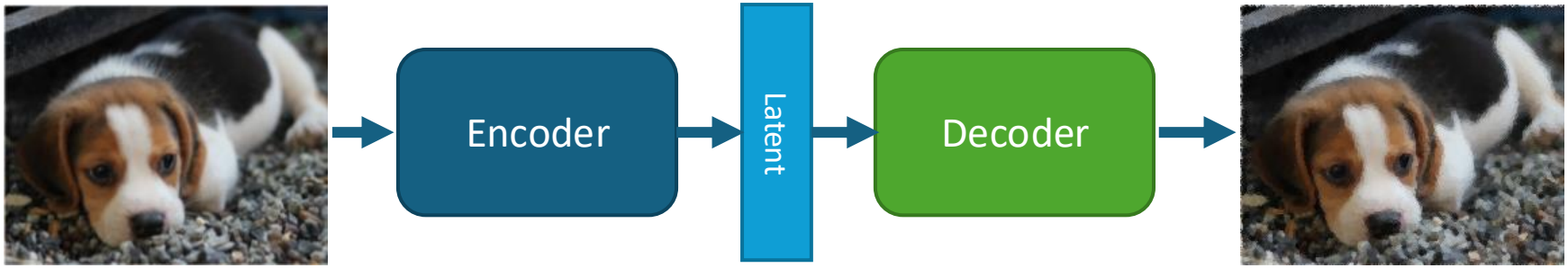
Assumptions: Normal instances can be better restructured from compressed space than anomalies



Methods: *Data reconstruction*

Assumptions: Normal instances can be better restructured from compressed space than anomalies

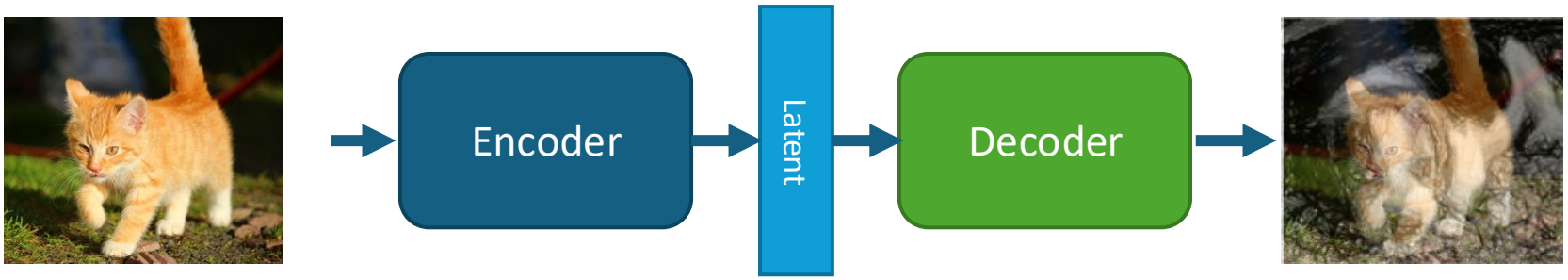
Testing



Methods: *Data reconstruction*

Assumptions: Normal instances can be better restructured from compressed space than anomalies

Testing

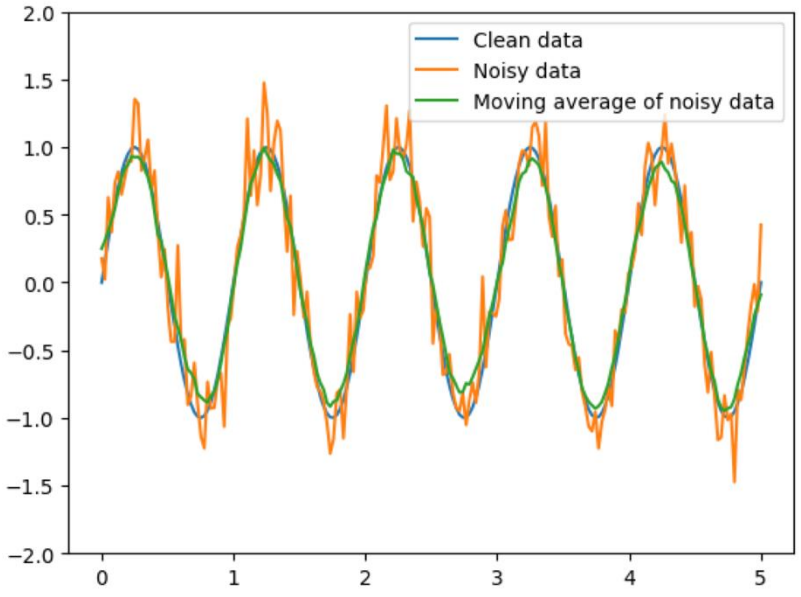


Anomaly Detection Techniques

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Time-Series Anomaly Detection

Moving average: compute the mean value of data within a window



For a time series x_t:

$$\hat{x}_t = \frac{1}{k} \sum_{i=t-k+1}^{t} x_i$$

k :window size

$\hat{x}_t$:expected (smoothed) value

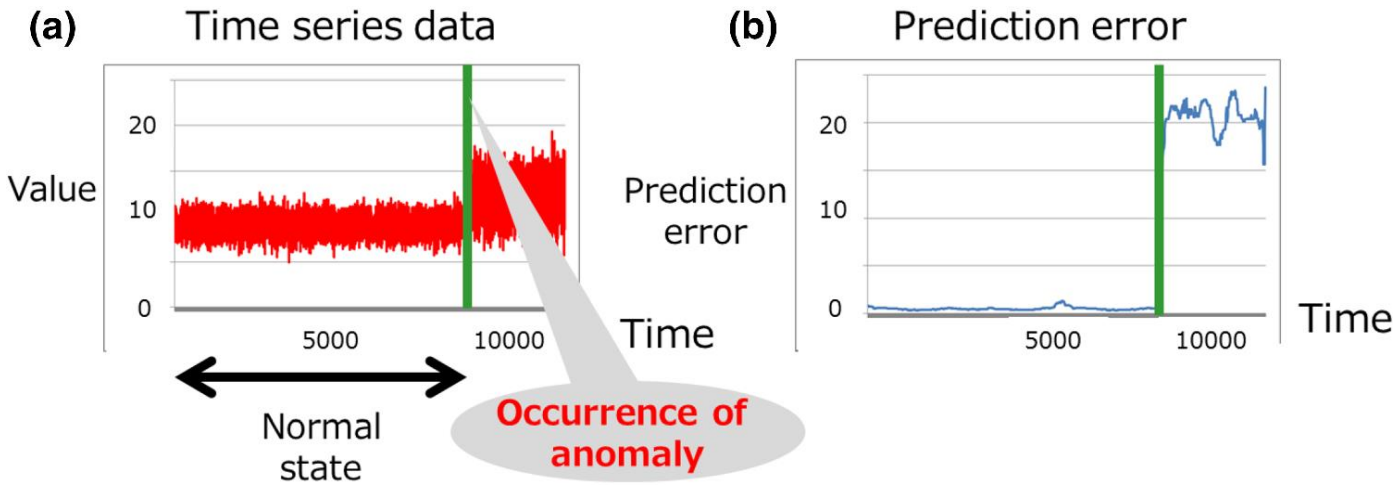
Compute Residual (Error)

$$e_t = x_t - \hat{x}_t$$

This measures how much the current point deviates from recent history

Time Series: *Predictability Modeling*

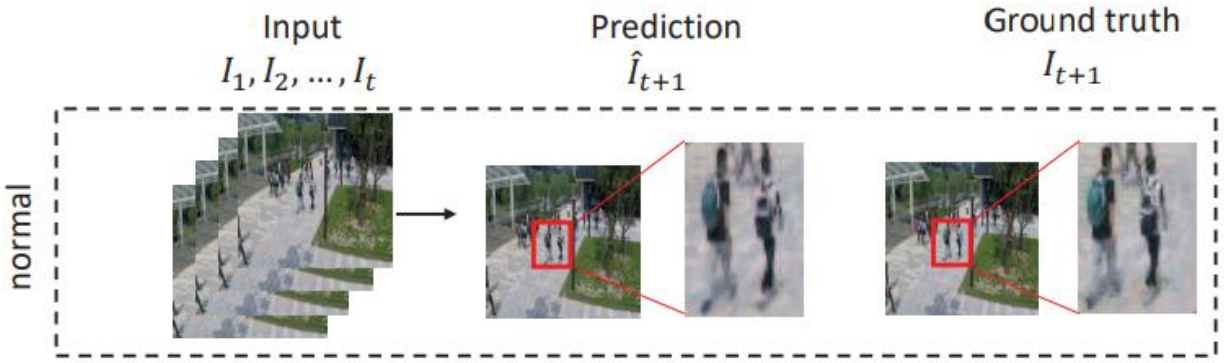
Assumptions: Normal instances are temporally more predictable than anomalies.



Maya S, Ueno K, Nishikawa T. dLSTM: a new approach for anomaly detection using deep learning with delayed prediction. International Journal of Data Science and Analytics. 2019 Sep;8(2):137-64.

Video: *Predictability Modeling*

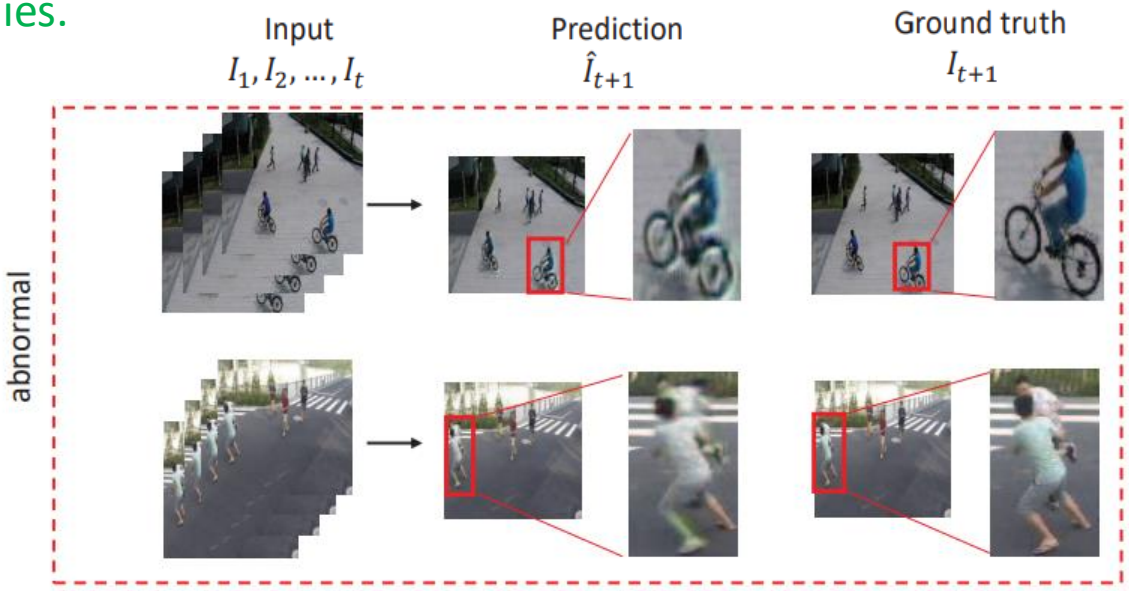
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Liu W, Luo W, Lian D, Gao S. Future frame prediction for anomaly detection–a new baseline. CVPR 2018

video: *Predictability Modeling*

Assumptions: Normal instances are temporally more predictable than anomalies.



Liu W, Luo W, Lian D, Gao S. Future frame prediction for anomaly detection–a new baseline. CVPR 2018

Strengths and Weaknesses

Core Idea: Learn a compact representation of normal data and detect anomalies via **reconstruction error**

Strengths

- **No distributional assumptions**
  → does not require modeling the underlying data distribution
- **Flexible modeling choices**
  → can use various dimensionality reduction methods (e.g., PCA, autoencoders)
- **Interpretable anomaly signal**
  → reconstruction error directly reflects deviation from normal patterns

Weaknesses

- **Sensitive to high dimensionality**
  → reconstruction error is computed in the original space
- **Error may become less informative in high dimensions**
  → small per-dimension errors can accumulate
- **Risk of over-generalization**
  → powerful models (e.g., deep autoencoders) may reconstruct anomalies well

Evaluation of Outlier Detection

- How do we measure performance?
- Why is evaluation challenging?

Why Evaluation is Difficult

Key Challenges

- Lack of labeled anomalies
- Severe class imbalance
- Ambiguity in ground truth
- Different definitions of “anomaly”

Evaluation depends heavily on available labels and application context

Evaluation Settings

Three Common Scenarios

- Fully Supervised
  - Ground truth labels available
- Semi-Supervised
  - Only normal data for training
  - Test set may contain anomalies
- Unsupervised
  - No labels at all

Different settings → different evaluation strategies

How to Evaluate Outlier Detector?

- If class labels are present
  - then use standard evaluation approaches for rare class such as precision, recall, or false positive rate
- If you don't have ground truth labels:
  - unsupervised evaluation:
    - use measures provided by the anomaly method, i.e., reconstruction error
    - Visual inspection: scatter plots (2D), PCA plots, t-SNE
  - Synthetic evaluation:
    - Manually inject anomalies into the data and test how well the detector finds them.
    - Useful for validating models before using them on real-world unlabeled data.

Summary of Outlier Detection

Goal: Detect rare or abnormal samples that do not conform to expected behavior.

Key assumption: Normal data is more frequent and follows a regular pattern; anomalies are rare and different.

Main categories of methods:

- Statistical methods: Assume a distribution (e.g., Gaussian) and flag low-probability points.
- Distance-based methods: Outliers are far from most other points (e.g., k-NN distance).
- Density-based methods: Outliers lie in low-density regions (e.g., LOF).
- Reconstruction-based methods: Outliers have high reconstruction error (e.g., PCA, autoencoders).
- Model-based methods: Learn boundary of normal data (e.g., One-Class SVM).

Summary

Strengths:

- Often works without labeled anomaly data
- Applicable to high-dimensional data (especially reconstruction-based)
- Flexible across domains (fraud, healthcare, sensor data)

Limitations:

- ❖ Sensitive to noise and hyperparameters
- ❖ High-dimensional data can degrade distance/density methods
- ❖ Assumptions about “normality” may be incorrect

Evaluation challenges:

- ➢ Anomalies are rare → imbalanced data
- ➢ Labels often incomplete or unavailable
- ➢ Metrics: Precision/Recall, F1, ROC-AUC, PR-AUC (preferred for imbalance)